Exercise Sheet 3

Discrete Mathematics by Hongfei Fu, 2023.9.18 & 2023.9.25

Name:Chen Zhuoyan   Student ID:523030910037   Class: 2302

Note: For the problems to check logical equivalence, if yes, please provide a rigorous proof; if no, please provide concrete domain and predicates that distinguish the truth values between the two predicate formulas.

1. Let $C(x)$ be the statement "x has a cat," let $D(x)$ be the statement "x has a dog," and let $F(x)$ be the statement "x has a ferret." Express each of these statements in terms of $C(x)$, $D(x)$, $F(x)$, quantifiers, and logical connectives. Let the domain consist of all students in your class.

   a) A student in your class has a cat, a dog, and a ferret.

   b) All students in your class have a cat, a dog, or a ferret.

   c) Some student in your class has a cat and a ferret, but not a dog.

   d) No student in your class has a cat, a dog, and a ferret.

   e) For each of the three animals, cats, dogs, and ferrets, there is a student in your class who has this animal as a pet.

   Answer Area:

   a) $\exists x(C(x) \land D(x) \land F(x))$.

   b) $\forall x(C(x) \lor D(x) \lor F(x))$.

   c) $\exists x(C(x) \land (\neg D(x)) \land F(x))$.

   d) $\forall x((\neg C(x)) \lor (\neg D(x)) \lor (\neg F(x)))$.

   e) $\exists x C(x) \lor \exists x D(x) \lor \exists x F(x)$.

2. Determine the truth value of each of these statements if the domain consists of all real numbers.

   a) $\exists x(x^3 = -1)$ b) $\exists x(x^4 < x^2)$
   c) $\forall x((-x)^2 = x^2)$ d) $\forall x(2x > x)$

   Answer Area:

   a) T b) T
   c) T d) F

3. Suppose the domain of the propositional function $P(x, y)$ consists of pairs x and y, where x is 1, 2, or 3 and y is 1, 2, or 3. Write out these propositions using disjunctions and conjunctions.

   a) $\exists x\ P(x, 3)$ d) $\forall x\ \neg P(x, 2)$

   Answer Area:
   a) $P(1, 3) \lor P(2, 3) \lor P(3, 3)$
   d) $(\neg P(1, 2)) \land (\neg P(2, 2)) \land (\neg P(3, 2))$

4. Determine whether $\forall x(P(x) \to Q(x))$ and $(\forall x P(x)) \to (\forall x Q(x))$ are logically equivalent. Furthermore, determine whether the following are true no matter what the domain is and what concrete predicate P and Q are:

   - $[\forall x(P(x) \to Q(x))] \to [(\forall x P(x)) \to (\forall x Q(x))]$
   - $[(\forall x P(x)) \to (\forall x Q(x))] \to [\forall x(P(x) \to Q(x))]$

   Please justify your answer.

   Addition: All variables have the same domain.

   Answer Area:
   $\forall x(P(x) \to Q(x))$ and $(\forall x P(x) \to (\forall x Q(x))$ are not logically equivalent.For example:
   $P(x) := x > 1$, $Q(x) : x^2 > 4$. Then domain of x is R
   $\because$ When $x = 1$, $P(x) \to Q(x)$ is false.
   $\therefore \forall x(P(x) \to Q(x))$ is false.
   $\because \forall x P(x)$ is false.
   $\therefore (\forall x P(x) \to (\forall x Q(x))$ is true.
   $\therefore$ The truth value of $\forall x(P(x) \to Q(x))$ and $(\forall x P(x) \to (\forall x Q(x))$ are not the same.
   $\therefore \forall x(P(x) \to Q(x))$ and $(\forall x P(x) \to (\forall x Q(x))$ are not logically equivalent.
   Furthermore:
   $\because$ Suppose that $\forall x(P(x) \to Q(x))$ is true.We clarify two cases.

   - For all x, P(x) is true.
     $\therefore$ For all x, $Q(x)$ is true.
     $\therefore \forall x P(x)$ is true,and $\forall x Q(x)$ is true.
     $\therefore (\forall x P(x) \to (\forall x Q(x))$ is true.
   - For some x,P(x) is false.
     $\therefore \forall x P(x)$ is false.
     $\therefore (\forall x P(x) \to (\forall x Q(x))$ is true.

   $\therefore [\forall x(P(x) \to Q(x))] \to [(\forall x P(x)) \to (\forall x Q(x))]$ is always true.
   $\because$ As is mentioned above, when $(\forall x P(x)) \to (\forall x Q(x))$ is true,$\forall x(P(x) \to Q(x))$ can be false.
   $\therefore [(\forall x P(x)) \to (\forall x Q(x))] \to [\forall x(P(x) \to Q(x))]$ isn't always true.

5. Determine whether $\forall a \forall b(P(a, b) \to P(b, a))$ and $\forall a \forall b(P(a, b) \leftrightarrow P(b, a))$ are logically equivalent, where the domains of the variables a, b are the same. Please justify your answer.

   Answer Area:
   A proof for $\forall a \forall b(P(a, b) \to P(b, a))$ and $\forall a \forall b(P(a, b) \leftrightarrow P(b, a))$ are logically equivalent is as follows.

   - Suppose that $\forall a \forall b(P(a, b) \to P(b, a))$ is true.
     $\because$ For an arbitrary $a = a^*$ and an arbitrary $b = b^*$ in the domain,$P(a^*, b^*) \to P(b^*, a^*)$ is true.
     $\therefore$ If we exchange the value of a and b, then $P(b^*, a^*) \to P(a^*, b^*)$ is true.
     $\therefore (\forall a \forall b(P(a, b) \to P(b, a))) \wedge (\forall a \forall b(P(b, a) \to P(b, a)))$ is true.
     $\therefore \forall a \forall b(P(a, b) \leftrightarrow P(b, a))$ is true.
   - Suppose that $\forall a \forall b(P(a, b) \leftrightarrow P(b, a))$ is true.
     $\therefore$ For all a and b,$P(a, b)$ and $P(b, a)$ are both true or both false.

$\therefore \forall a \forall b(P(a,b) \to P(b,a))$ is true.

In conclusion, $\forall a \forall b(P(a,b) \to P(b,a))$ and $\forall a \forall b(P(a,b) \leftrightarrow P(b,a))$ are logically equivalent.

6. Establish these logical equivalences, where x does not occur as a free variable in A. Assume that the domain is nonempt.
a) $\forall x(A \to P(x)) \equiv A \to \forall x P(x)$
b) $\exists x(A \to P(x)) \equiv A \to \exists x P(x)$
Note: In Exercise 48, we assume that the statement A does not have free variables and is either true or false but not both.
Answer Area:
a) A proof for $\forall x(A \to P(x)) \equiv A \to \forall x P(x)$ is as follows.

- Suppose that $\forall x(A \to P(x))$ is true.We clarify two cases.
  - A is false.Hence, $A \to \forall x P(x)$ is true.
  - A is true.Then for all x in the domain, $P(x)$ is true. Hence, $\forall x P(x)$ is true.Therefore, $A \to \forall x P(x)$ is true.
- Suppose that $A \to \forall x P(x)$ is true.We clarify two cases.
  - A is false.Then for all x in the domain, $A \to P(x)$ is true. Hence, $\forall x(A \to P(x))$ is true.
  - A is true. Then $\forall x P(x)$ is true. Hence, for all x in the domain, $P(x)$ is true, $A \to P(x)$ is true. Therefore, $\forall x(A \to P(x))$ is true.

In conclusion, $\forall x(A \to P(x)) \equiv A \to \forall x P(x)$.

b) A proof for $\exists x(A \to P(x)) \equiv A \to \exists x P(x)$ is as follows.

- Suppose that $\exists x(A \to P(x))$ is true.We clarify two cases.
  - A is false.Hence, $A \to \exists x P(x)$ is true.
  - A is true.Then there is a x^* in the domain, $P(x^8)$ is true. Hence, $\exists x P(x)$ is true.Therefore, $A \to \exists x P(x)$ is true.
- Suppose that $A \to \exists x P(x)$ is true.We clarify two cases.
  - A is false.Then for all x in the domain, $A \to P(x)$ is true. We can choose an arbitrary x^* in the domain and ensure that $A \to P(x^*)$ is true. Hence, $\exists x(A \to P(x))$ is true.
  - A is true. Then $\exists x P(x)$ is true. Then there is a x^* such that $P(x^*)$ is true, and $A \to P(x^*)$ is true. Therefore, $\exists x(A \to P(x))$ is true.

In conclusion, $\forall x(A \to P(x)) \equiv A \to \forall x P(x)$.

7. Determine these logical equivalences, where x does not occur as a free variable in A. Assume that the domain is nonempt.
a) $\forall x(P(x) \to A) \equiv (\forall x P(x)) \to A$
b) $\exists x(P(x) \to A) \equiv (\exists x P(x)) \to A$
Note: We assume that the statement A does not have free variables and is either true or false but not both.
Answer Area:

a) $\forall x(P(x) \to A)$ and $(\forall x P(x)) \to A$ are not logically equivalent. For example:
A is false, and $P(x) := x > 1$.The domain of x is R
$\because$ When $x = 2$, $P(x)$ is true, and A is false.
$\therefore P(x) \to A$ is false.
$\therefore \forall x(P(x) \to A)$ is false.
$\because$ When $x = 0$, $P(x)$ is false.
$\therefore \forall x P(x)$ is false.
$\therefore (\forall x P(x)) \to A$ is true.
$\therefore \forall x(P(x) \to A)$ and $(\forall x P(x)) \to A$ are not logically equivalent.

b)$\exists x(P(x) \to A)$ and $(\exists x P(x)) \to A$ are not logically equivalent.For example:
A is false, and $P(x) := x > 1$.The domain of x is R
$\because$ When $x = 0$, $P(x)$ is false, and A is false.
$\therefore P(x) \to A$ is true.
$\therefore \exists x(P(x) \to A)$ is true.
$\because$ When $x = 2$, $P(x)$ is true.
$\therefore \exists x P(x)$ is true.
$\therefore (\exists x P(x)) \to A$ is false.
$\therefore \exists x(P(x) \to A)$ and $(\exists x P(x)) \to A$ are not logically equivalent.

8. Show that $(\forall x P(x)) \vee (\forall x Q(x))$ and $\forall x(P(x) \vee Q(x))$ are not logically equivalent.

Answer Area:
Suppose that:
$P(x) := x \geqslant 1$, $Q(x) := x < 1$.The domain consist of all real numbers.
$\because \forall x \in R$,$x \geqslant 1$ or $x < 1$.
$\therefore \forall x(P(x) \vee Q(x))$ is true.
$\because (\forall x P(x))$ is false and $(\forall x Q(x))$ is false too.
$\therefore (\forall x P(x)) \vee (\forall x Q(x))$ is false.
$\therefore (\forall x P(x)) \vee (\forall x Q(x))$ and $\forall x(P(x) \vee Q(x))$ are not logically equivalent.

9. Show that $(\exists x P(x)) \wedge (\exists x Q(x))$ and $\exists x(P(x) \wedge Q(x))$ are not logically equivalent.

Answer Area:
Suppose that:
$P(x) := x > 1$, $Q(x) := x < 1$.The domain consist of all real numbers.
$\because \exists x \in R$,$x > 1$,and $\exists x \in R$,$x < 1$
$\therefore (\exists x P(x)) \wedge (\exists x Q(x))$ is true.
$\because \forall x$ $(x > 1)$ and $(x < 1)$ can't be true at the same time.
$\therefore \exists x(P(x) \wedge Q(x))$ is false.
$\therefore (\exists x P(x)) \wedge (\exists x Q(x))$ and $\exists x(P(x) \wedge Q(x))$ are not logically equivalent.

10. Use predicates, quantifiers, logical connectives, and mathematical operators to express the statement that there is a positive integer that is not the sum of three squares.
Note: In Exercise 22 we fix the domain for all variables to be integers. We also assume that a square here is a square of some integer (i.e., not real number).

Answer Area:
Define P(x,a,b,c):= $x = a^2 + b^2 + c^2$.
$x \in Z^+$. $a, b, c \in Z$.
$\exists x \forall a \forall b \forall c P(x, a, b, c)$

11. Express the negations of each of these statements so that all negation symbols immediately precede predicates.
a) $\exists z \forall y \forall x T(x, y, z)$
b) $(\exists x \exists y P(x, y)) \wedge (\forall x \forall y Q(x, y))$
c) $\exists x \exists y (Q(x, y) \leftrightarrow Q(y, x))$
d) $\forall y \exists x \exists z (T(x, y, z) \vee Q(x, y))$

Answer Area:

a) $\forall z \exists y \exists x \neg T(x, y, z)$
b) $(\forall x \forall y \neg P(x, y)) \vee (\exists x \exists y \neg Q(x, y))$
c) $\forall x \forall y ((\neg P(x, y) \vee \neg Q(x, y)) \wedge (P(x, y) \vee Q(x, y)))$
d) $\exists y \forall x \forall z (\neg T(x, y, z) \wedge \neg Q(x, y))$

12. Show that $(\forall x P(x)) \wedge (\exists x Q(x))$ is logically equivalent to $\forall x \exists y (P(x) \wedge Q(y))$, where all quantifiers have the same nonempty domain.

Answer Area:
A proof for $(\forall x P(x)) \wedge (\exists x Q(x))$ is logically equivalent to $\forall x \exists y (P(x) \wedge Q(y))$ is as follows.

- Suppose that $(\forall x P(x)) \wedge (\exists x Q(x))$ is true.
  $\therefore$ $\forall x P(x)$ is true, and $\exists x Q(x)$ is true.
  $\therefore$ $\exists y Q(y)$ is true.
  $\therefore$ For all x in the domain,there is a y such that $Q(y)$ is true.That is $\exists y (P(x) \wedge Q(y))$ is true.
  $\therefore$ $\forall x \exists y (P(x) \wedge Q(y))$ is true.
- Suppose that $\forall x \exists y (P(x) \wedge Q(y))$ is true.
  $\therefore$ $(\forall x P(x))$ and $(\exists y Q(y))$ are both true.
  $\therefore$ $(\forall x P(x)) \wedge (\exists x Q(x))$ is true.

13. Determine whether $(\exists x\, P(x)) \to (\exists x\, Q(x))$ and $\exists x \exists y\, (P(x) \to Q(y))$ are logically equivalent. Assume the variables x,y have the same non-empty domain. (2 points)

Answer Area:
$(\exists x\, P(x)) \to (\exists x\, Q(x))$ and $\exists x \exists y\, (P(x) \to Q(y))$ are not logically equivalent. For example: $P(x) := x > 1$, $Q(x) := x^2 < 0$. The domain consists of all real numbers.
$\because$ $\exists x P(x)$ is true.And for all x in the domain, $Q(x)$ is false.
$\therefore$ $\exists x Q(x)$ is false.Then $(\exists x\, P(x)) \to (\exists x\, Q(x))$ is false.
$\because$ When $x = 0, y = 0$, $P(x)$ is false and $Q(y)$ is false. Then $(P(x) \to Q(y))$ is true.
$\therefore$ $\exists x \exists y\, (P(x) \to Q(y))$ is true.
$\therefore$ $(\exists x\, P(x)) \to (\exists x\, Q(x))$ and $\exists x \exists y\, (P(x) \to Q(y))$ aren't logically equivalent.

14. Express the quantification $\exists! x P(x)$, introduced in Section 1.4, using universal quantifications, existential quantifications, and logical operators.

    Answer Area:
    $\exists x P(x) \wedge (\forall y((y \neq x) \rightarrow \neg P(y)))$

15. Determine whether the two following predicate formulas are logically equivalent and write your reasons. (2 points)

    - $\exists x\, (\forall y\, (P(y) \leftrightarrow y = x))$
    - $\exists x\, (\forall y\, (P(y) \rightarrow y = x))$

    Answer Area:
    $\exists x\, (\forall y\, (P(y) \leftrightarrow y = x))$ and $\exists x\, (\forall y\, (P(y) \rightarrow y = x))$ are not logically equivalent.
    For example:
    $P(y) := y^2 < 0$. The domain consists of all real numbers.
    $\because$ For all y in the domain, $P(y)$ is false.
    $\therefore \forall y\, (P(y) \rightarrow y = x)$ is true.
    $\therefore$ We can choose an arbitrary x^* in the domain and ensure that $\forall y\, (P(y) \rightarrow y = x^*)$ is true.
    $\therefore \exists x\, (\forall y\, (P(y) \rightarrow y = x))$ is true.
    $\because$ There is an arbitrary x^* in the domain, when $y = x^*$, $P(y)$ is false, and $y = x$ is true. Then $P(y) \rightarrow y = x$ is false.
    $\therefore \forall y\, (P(y) \leftrightarrow y = x^*)$ is false.
    $\therefore \exists x\, (\forall y\, (P(y) \leftrightarrow y = x))$ is false.
    In conclusion, $\exists x\, (\forall y\, (P(y) \leftrightarrow y = x))$ and $\exists x\, (\forall y\, (P(y) \rightarrow y = x))$ are not logically equivalent.